COOLSHADE · PROJECT PLAN

CoolShade project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/11-plan.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	First issue in the kit 1.9.1 upgrade: CoolShade is at stage P3 (design outputs), TRL 3; P4 to P6 on hold under the stage cap

CoolShade project plan

CoolShade is at stage P3 (design outputs), TRL 3. The calculation note, the D3 model of all three concept options, the drawings, the priced BOM and the build plan are done; the concept choice is proposed and waits for Amish (CSH-DDR-004). The next gate, P4 (verification, TRL 4), is on hold while the portfolio stage cap is P3, and needs a site and a hygiene plan first. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept choice	Amish Chadha	CSH-OPT-001, CSH-DDR-004, CSH-DEC-001 D8	Proposed, awaiting Amish
Partner and site	Amish Chadha	Market operator and a heat-research partner in a hot, dry city (CSH-DEC-001)	Candidates named; nothing agreed
Requirements	Design lead	CSH-URS-001, CSH-PRS-001, CSH-RTM-001	Done on paper; partner review pending
STEM check	Design lead	CSH-STM-001, `stem_check.py`	Done 2026-10-30; 13 rows plausible, need field data
Calculations and model	Design lead	CSH-CAL-001 v0.6; D3 model of options A, B and C; STEP and STL	Done 2026-10-30
Drawings, BOM, build plan	Design lead	CSH-DWG-001, DWG-101 to 116, `bom/bom.csv`, CSH-BLD-001	Done 2026-09-30, redrawn 2026-10-30
Risk, market, regulatory	Design lead	CSH-RSK-001 with design FMEA, CSH-MKT-001, CSH-REG-001	Done 2026-10-30
Renders	Coordinator	Options picture and product renders (Blender, Amish's Mac)	Queued after the kit 1.9.1 upgrade
Verification (P4)	Partner	Hygiene sampling, nozzle spectrum, build and bench tests	On hold (stage cap P3)

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Design lead (Claude with Amish's review)	P0 to P3 documents	Done	0
Prototype parts	One canopy (P4, on hold)		882 (`bom/bom.csv`, 107 over the 775 target)
Nozzle maker's data	Flow and Dv0.9 at 7 bar	A request	To assess
Water hygiene plan	Plan accepted by the health authority	Weeks, with the market operator	To assess
Site survey	Slab, wind, setbacks, water hardness	1 to 2 days	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Done 2026-10-30 (kit 1.9.1 upgrade of a repo made at kit 1.7)
P1 Definition	URS, PRS draft, RSK, MKT, REG complete	Done 2026-10-30
P2 Design inputs	PRS targets, CAL, RTM, DDR, STEM check closed by calculation	Done 2026-10-30; 13 STEM rows need field data
P3 Design outputs	Model, STEP, DWG, BOM, BLD, DEC, FMEA	Done 2026-10-30
P4 Verification	Hygiene sampling, nozzle test, bench and wind-load checks	On hold
P5 to P6	Field trials with a partner, launch	On hold

Next steps and decision requested

Decision requested from Amish: the concept choice (option A, B or C, or a hybrid of A with option C's white skin), the presence sensor trade-off (PIR or radar) and whether to disinfect the tank water (CSH-DEC-001 D8, D10, D11). The recommended next step is a partner and a hygiene plan and, when the cap allows, TRL 4: the nozzle's droplet spectrum, one canopy built from the plan, and a bench test of the controller.

References

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